

Unit 13a: Earth's Interior, Continental Drift & Seafloor Spreading

Reading: pages 179–190 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. S-waves cannot travel through liquid. Their disappearance at the mantle–core boundary tells scientists that
 - a) the outer core is liquid
 - b) the inner core is liquid
 - c) the crust is unusually thin
 - d) the mantle is liquid
2. Which pair of layers divides Earth by *mechanical* properties rather than by composition?
 - a) Oceanic crust and continental crust
 - b) Inner core and outer core
 - c) Core, mantle, and crust
 - d) Lithosphere and asthenosphere
3. Approximately what share of Earth's mass is the mantle?
 - a) Less than 1%
 - b) About 31%
 - c) About 68%
 - d) About 85%
4. Who proposed the continental drift hypothesis and named the supercontinent Pangaea?
 - a) Alfred Wegener
 - b) James Hutton
 - c) William Smith
 - d) Nicholas Steno
5. Why did most geologists reject Wegener's hypothesis during his lifetime?
 - a) His fossil evidence turned out to be fabricated
 - b) There was no way to explain how solid continents could plow through solid oceanic crust
 - c) The continents plainly do not fit together
 - d) The magnetic evidence directly contradicted it

6. The oldest seafloor anywhere on Earth is less than about
- a) 18 million years old
 - b) 180 million years old
 - c) 1.8 billion years old
 - d) 4.5 billion years old

Part B · True or False

2 questions × 1 point = 2 points

7. P-waves travel faster than S-waves and arrive at a seismograph first.

TRUE / FALSE

8. Continental crust is denser than oceanic crust.

TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Describe three separate lines of evidence Wegener used to support continental drift.

10. Explain how the pattern of magnetic striping on the seafloor supports the seafloor spreading hypothesis.

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